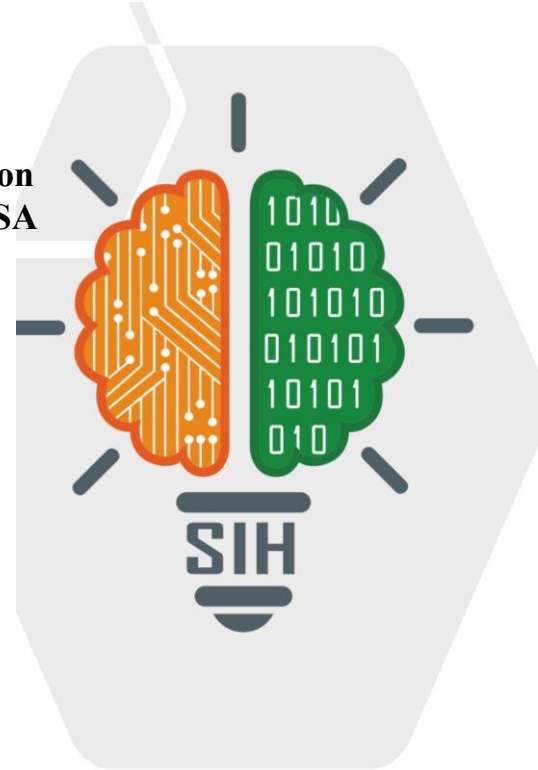


SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- **Problem Statement ID - SIH26162**
- **Problem Statement Title- AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal) - Fire Orbit**



Organisation: National Technical Research Organisation (NTRO)

ThermalSentinel: Classifying Every Satellite Hotspot and Alerting Only on Abnormal Industrial Heat



Real-world issue: NASA FIRMS reports thousands of hot pixels over India every day. A refinery flare, a steel plant, a coal-seam fire, a stubble burn and a forest fire all look identical in that feed.

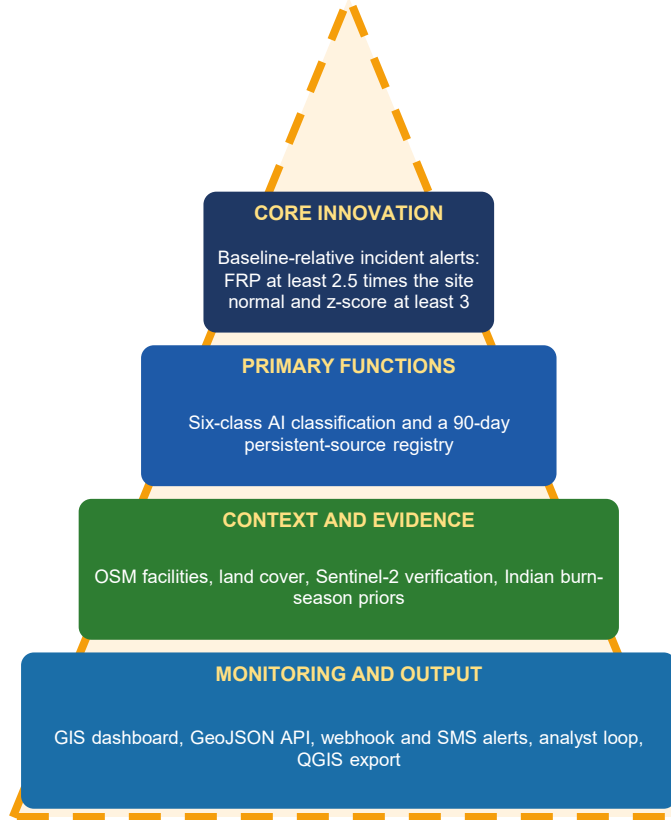
Why it matters: A genuine plant fire hides among routine industrial heat, so responders cannot prioritise, and no registry of persistent thermal sources exists for security agencies.

Our solution: Classify every hotspot into 6 classes, track each source for 90 days, and raise an alert only when a known site burns far above its own normal level.

[Live Demo](#)
[GitHub](#)

Prototype

running today, click to open



Risk

vs

Solution

Identical hotspots carry no meaning



Six-class AI: industrial, flare, mine, crop, wildfire

Real fires hidden in routine heat



Alert only when FRP exceeds the site baseline

No memory of persistent sources



90-day source tracking and registry

Black-box AI is not trusted



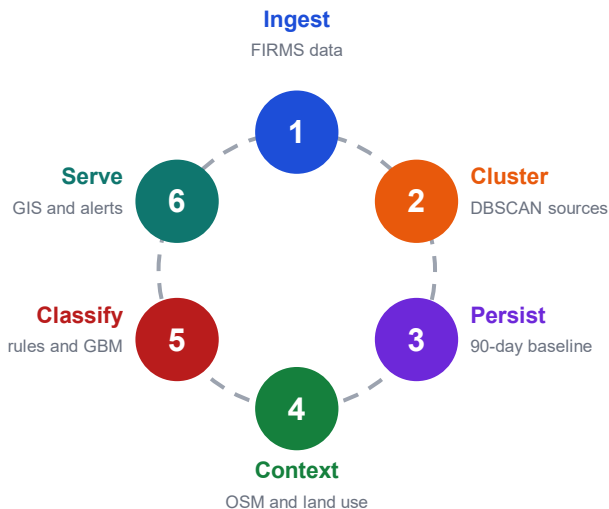
Every label carries plain reasons



Prototype GIS dashboard

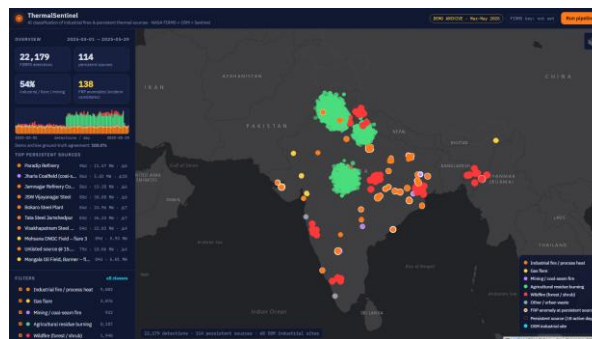
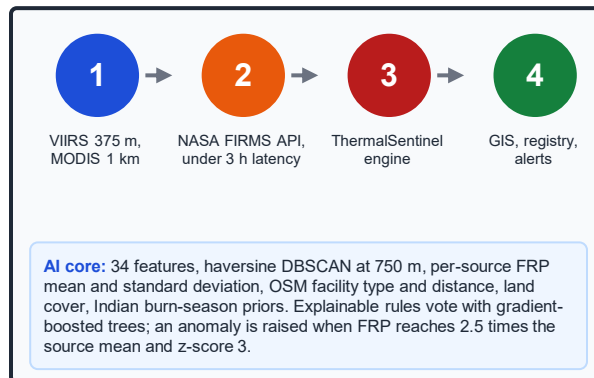
TECHNICAL APPROACH

METHODOLOGY AND PROCESS OF IMPLEMENTATION



[Report
GitHub](#)

Detailed
Report



Prototype: 22,179 detections classified, 114 persistent sources, 138 incident alerts, GeoJSON export

TECHNOLOGIES USED

1 Data
NASA FIRMS (VIIRS 375 m, MODIS), OpenStreetMap Overpass, Copernicus Sentinel-2, NASA Worldview

2 AI and ML
Python, scikit-learn (DBSCAN, HistGradientBoosting), rule engine, shapely STRtree

3 Backend
FastAPI REST with GeoJSON, SQLite moving to PostGIS, OGC-standard exports

4 Frontend
React 18 with Vite, Leaflet canvas renderer for 40,000+ points, deep-link views

5 Deployment
GitHub Pages demo, single Linux VM, NIC cloud or air-gapped server, Docker-ready



FEASIBILITY ANALYSIS

Working prototype: the full 90-day archive of 22,179 detections is classified in about 7 seconds on a laptop, and the demo is public.

Open data only: the FIRMS key is issued instantly, OSM Overpass and Copernicus are free. No licences, no procurement.

Light footprint: FastAPI, SQLite and React run on a single VM, NIC cloud or an air-gapped NTRO server.

One switch to live: a single environment variable moves the same pipeline from the archive to near-real-time FIRMS.

VIABILITY

Signals that work: persistence, OSM context and physical signature separate the classes. Accuracy on the labelled 90-day archive is 99.9 % hybrid and 98.4 % rules-only.

Designed for trust: every label shows its reasons. Analysts confirm or override, and their labels retrain the model.

Scales: OSM is queried only around hotspot cells, with a PostGIS and tile path for national, multi-year archives.

Risks handled: OSM gaps use a curated list and WorldCover, clouds are covered by a 90-day window across 3 satellites, false alarms by baseline-relative thresholds.

BUSINESS AND DEPLOYMENT

Government: SDMA control rooms, an NTRO source registry and CPCB fire attribution, at roughly Rs 3,000 to 5,000 per month per deployment.

Industry and insurers: site-monitoring subscriptions and verified incident evidence for claims and compliance.

Integration: GeoJSON layers for Bhuvan, NDEM, QGIS and ArcGIS. No new GIS is required.

Roadmap: a 4-week live pilot with validation on real incidents, then alerting and a Sentinel-2 chip classifier, then an agency SOP.

Idea and prototype

done, public demo live

Next 4 weeks

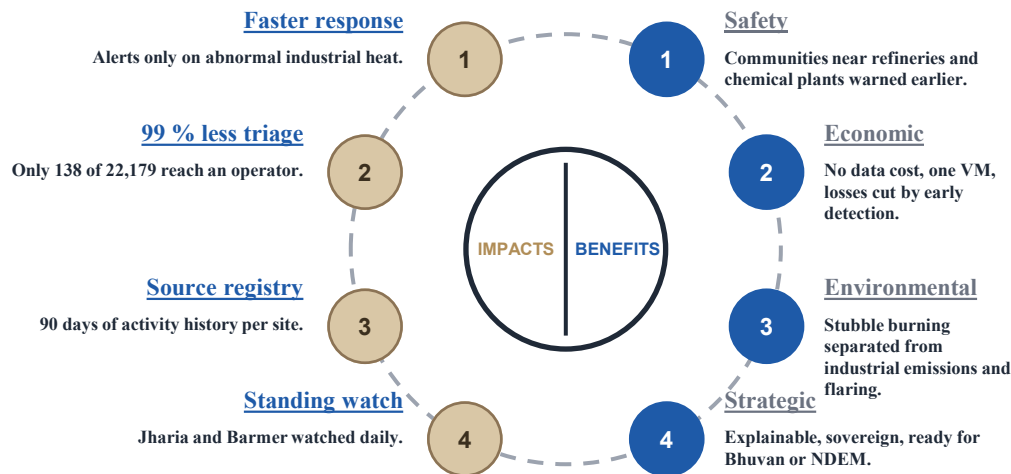
live FIRMS pilot, validation

Grand finale

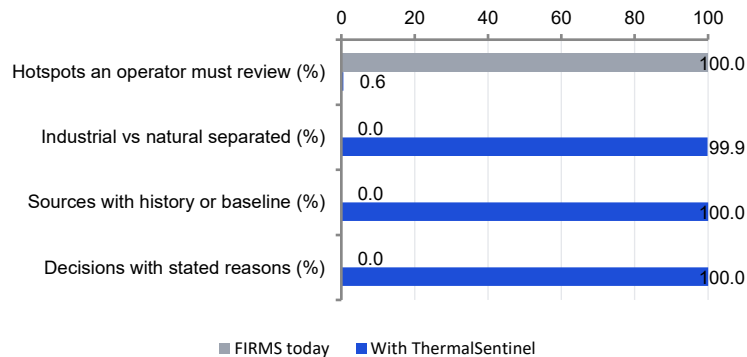
alerting, Sentinel-2 classifier

Deployment

Bhuvan and NDEM, SDMA SOP



Control-room workload: FIRMS today versus ThermalSentinel

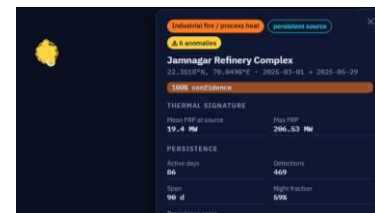


Measured on the labelled 90-day archive of 22,179 detections.

From hot pixels to actionable alerts



crop-burn season, Punjab



Jamnagar: 86 active days, 207 MW spike

policy and problem

hotspot data source

sensor science

flare signatures

industrial heat

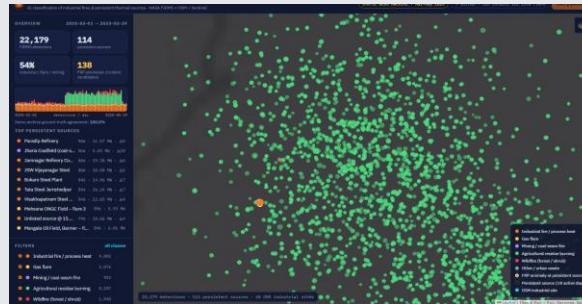
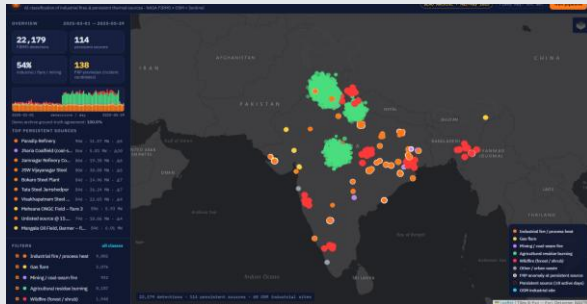
context data

verification

**NTRO
PS SIH26162
SIH 2026**
[LINK](#)
**NASA FIRMS
Active-fire
Area API**
[LINK](#)
**Schroeder 2014
VIIRS 375 m
fire product**
[LINK](#)
**Elvidge 2013
VIIRS
Nightfire**
[LINK](#)
**Liu 2018
Industrial heat
sources (RSE)**
[LINK](#)
**OpenStreetMap
Overpass API
and tags**
[LINK](#)
**Copernicus
Sentinel-2 SWIR
WorldCover**
[LINK](#)

Working prototype screens

all-India classification, Punjab crop-burn season, Jharia coal-seam fires

Working prototype: <https://rahulskandagal.github.io/thermal-sentinel/>Source code: <https://github.com/rahulskandagal/thermal-sentinel>